

# HW1

Greyson Knippel  
CPSC 321 — Gonzaga University  
compiled with L<sup>A</sup>T<sub>E</sub>X and TikZ :)

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## part A

### question 1

```
AIRPORT(air_id, air_name, city, state)
FK: none
```

`air_id` is the primary key since its unique, and minimal.

| <b>air_id</b> | <b>air_name</b>              | <b>city</b> | <b>state</b> |
|---------------|------------------------------|-------------|--------------|
| GEG           | Spokane International        | Spokane     | WA           |
| SEA           | Seattle-Tacoma International | Seattle     | WA           |
| LAX           | Los Angeles International    | Los Angeles | CA           |

### question 2

```
AIRLINE(air_code, airline_name, air_id, founded_year)
FK: air_id -> AIRPORT(air_id)
```

`air_code` is the primary key since its unique, and minimal. `air_id` is a foreign key referencing the hub airport's `air_id` in AIRPORT.

| <b>air_code</b> | <b>airline_name</b> | <b>air_id</b> | <b>founded_year</b> |
|-----------------|---------------------|---------------|---------------------|
| AS              | Alaska Airlines     | SEA           | 1932                |
| WN              | Southwest Airlines  | LAX           | 1967                |
| UA              | United Airlines     | GEG           | 1926                |

### question 3

```
FLIGHT_ROUTE(air_code, flight_num, dep_id, arr_id, distance)
FK: air_code -> AIRLINE(air_code)
FK: dep_id -> AIRPORT(arr_id)
FK: arr_id -> AIRPORT(arr_id)
```

primary key is composite. neither attribute is unique on its own: an airline operates many routes, so `air_code` repeats, and two airlines can both use flight number 122, so `flight_num` repeats. together they are unique, and removing either one breaks uniqueness, so the pair is minimal.

`dep_id` and `arr_id` are both foreign keys into `AIRPORT`.

| <u>air_code</u> | <u>flight_num</u> | <u>dep_id</u> | <u>arr_id</u> | <u>distance</u> |
|-----------------|-------------------|---------------|---------------|-----------------|
| AS              | 122               | GEG           | SEA           | 224             |
| AS              | 345               | SEA           | LAX           | 954             |
| WN              | 122               | LAX           | GEG           | 1046            |

### question 4

```
PASSENGER(pass_id, first_name, last_name, email)
Additional candidate key: {email}
FK: none
```

`pass_id` and `email` are both candidate keys since both are stated to be unique. `pass_id` should be the primary key because email addresses can change, and that would be bad.

| <u>pass_id</u> | <u>first_name</u> | <u>last_name</u> | <u>email</u>         |
|----------------|-------------------|------------------|----------------------|
| 1              | Jack              | Smith            | jsmith@example.com   |
| 2              | Alan              | Cho              | acho@example.com     |
| 3              | Greyson           | Knippel          | gknippel@example.com |

```
BOOKING(pass_id, air_code, flight_num, travel_date, seat, price)
FK: pass_id -> PASSENGER(pass_id)
FK: (air_code, flight_num) -> FLIGHT_ROUTE(air_code, flight_num)
```

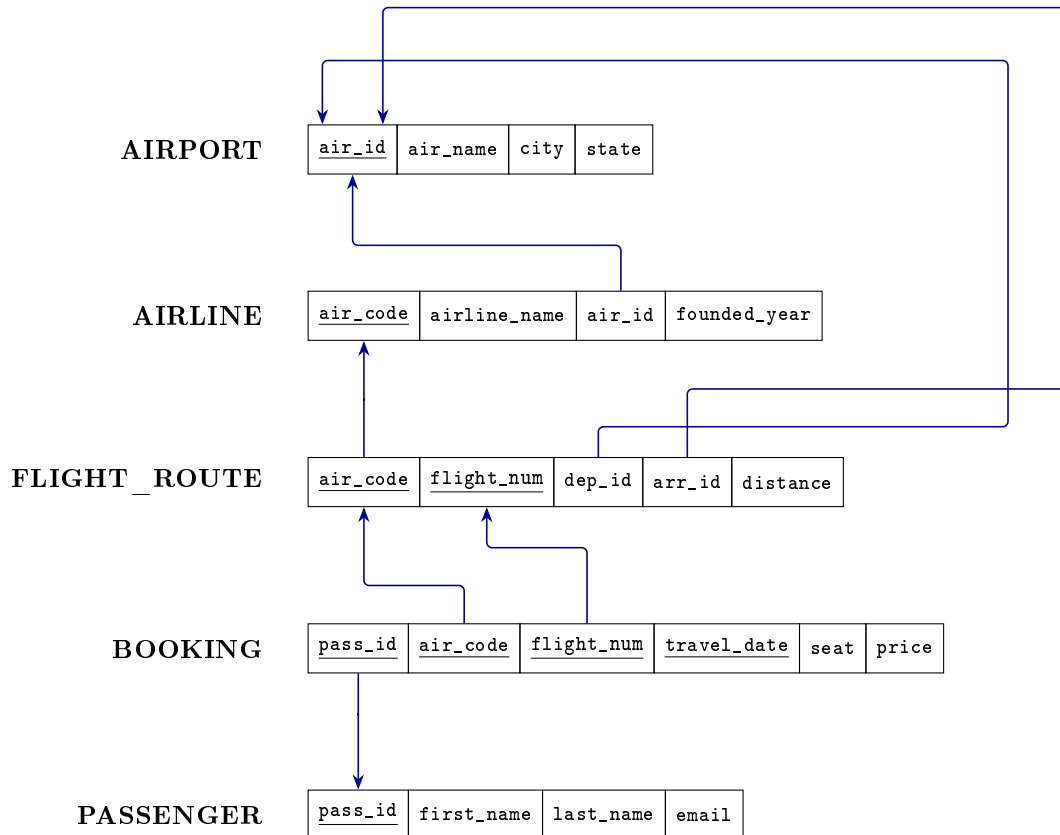
the constraint “booked once per route per date” makes those four attributes the key. every removal of the attributes breaks it as well.

(`air_code`, `flight_num`) is a composite foreign key the parent key is composite, so the reference has to carry both columns together. `seat` and `price` are irrelevant to this table’s keys.

| <u>pass_id</u> | <u>air_code</u> | <u>flight_num</u> | <u>travel_date</u> | seat | price  |
|----------------|-----------------|-------------------|--------------------|------|--------|
| 1              | AS              | 122               | 2026-10-02         | 14A  | 189.50 |
| 1              | AS              | 122               | 2026-11-15         | 08C  | 210.00 |
| 2              | AS              | 122               | 2026-10-02         | 14B  | 205.75 |

## part B

### question 5



the underlined attributes are primary keys. each arrow runs from a foreign key to the primary key it references. the two arrows from BOOKING into FLIGHT\_ROUTE are a single composite foreign key: air\_code and flight\_num must be carried together, since the key they reference is composite. AIRPORT is referenced three times: once by AIRLINE for its hub, and twice by FLIGHT\_ROUTE for the departure and arrival airports.

## part C

### question 6

(a)

```
MaintenanceRecord(aircraft_id, maintenance_date, mechanic_id,  
    aircraft_model, manufacturer, mechanic_name, work_performed,  
    parts_cost, mechanic_cert_level)  
FK: none
```

the key means one row is “this mechanic worked on this aircraft on this date,” so other attributes should relate to the situation.

the redundant values are `mechanic_name` and `mechanic_cert_level`. both are facts about the mechanic, not about a maintenance job, so they repeat on every record that mechanic performs:

| attribute                        | really a fact about          | so it repeats                  |
|----------------------------------|------------------------------|--------------------------------|
| <code>aircraft_model</code>      | the aircraft                 | every record for that aircraft |
| <code>manufacturer</code>        | the aircraft (and its model) | every record for that aircraft |
| <code>mechanic_name</code>       | the mechanic                 | every record by that mechanic  |
| <code>mechanic_cert_level</code> | the mechanic                 | every record by that mechanic  |
| <code>work_performed</code>      | the visit itself             | never                          |
| <code>parts_cost</code>          | the visit itself             | never                          |

a 737-900 is always built by Boeing, so the same value also repeats across completely different aircraft that happen to share a model.

only `work_performed` and `parts_cost` change from one maintenance visit to the next, so they are the only attributes that actually belong in this table.

| aircraft | date       | mech | model   | manuf. | mech_name   | work       | cost    | cert |
|----------|------------|------|---------|--------|-------------|------------|---------|------|
| N101AS   | 2026-03-04 | M7   | 737-900 | Boeing | James Smith | brake pads | 1450.00 | A&P  |
| N101AS   | 2026-06-18 | M7   | 737-900 | Boeing | James Smith | hydraulics | 0.00    | A&P  |
| N220WN   | 2026-06-18 | M7   | 737-700 | Boeing | James Smith | oil change | 310.00  | A&P  |
| N101AS   | 2026-09-02 | M3   | 737-900 | Boeing | James Smith | tires      | 2100.00 | IA   |

“James Smith” and “A&P” are written three times because M7 appears three times, and “Boeing” is written on all four rows despite two different aircraft and two different models.

(b)

each group of facts from part (a) moves into its own table, and the keys are whatever decides it. what is left behind in the original table is just the id, which links back to the new table:

```
AIRCRAFT_MODEL(aircraft_model, manufacturer)
FK: none

AIRCRAFT(aircraft_id, aircraft_model)
FK: aircraft_model -> AIRCRAFT_MODEL(aircraft_model)

MECHANIC(mechanic_id, mechanic_name, mechanic_cert_level)
FK: none

MAINTENANCE_RECORD(aircraft_id, maintenance_date, mechanic_id,
    work_performed, parts_cost)
FK: aircraft_id -> AIRCRAFT(aircraft_id)
FK: mechanic_id -> MECHANIC(mechanic_id)
```

the key of MAINTENANCE\_RECORD does not change: it is still the same three attributes but `aircraft_id` and `mechanic_id` are now foreign keys as well as key attributes. only `work_performed` and `parts_cost` stay behind, since those are the only values that actually describe the visit.

the fourth table is what removes the last of the redundancy. if `manufacturer` were left in AIRCRAFT, “Boeing” would still have to be written once per aircraft. giving the model its own table means each manufacturer is recorded exactly once.

| <u>aircraft_model</u> | manufacturer | <u>aircraft_id</u> | <u>aircraft_model</u> |
|-----------------------|--------------|--------------------|-----------------------|
| 737-900               | Boeing       | N101AS             | 737-900               |
| 737-700               | Boeing       | N220WN             | 737-700               |
| A320-200              | Airbus       | N455UA             | A320-200              |

| <u>mechanic_id</u> | <u>mechanic_name</u> | <u>mechanic_cert_level</u> |
|--------------------|----------------------|----------------------------|
| M7                 | Dana Weiss           | A&P                        |
| M3                 | Omar Diaz            | IA                         |
| M9                 | Lena Petrov          | A&P                        |

| <u>aircraft_id</u> | <u>maintenance_date</u> | <u>mechanic_id</u> | <u>work_performed</u> | <u>parts_cost</u> |
|--------------------|-------------------------|--------------------|-----------------------|-------------------|
| N101AS             | 2026-03-04              | M7                 | brake pads            | 1450.00           |
| N101AS             | 2026-06-18              | M7                 | hydraulics            | 0.00              |
| N220WN             | 2026-06-18              | M7                 | oil change            | 310.00            |
| N101AS             | 2026-09-02              | M3                 | tires                 | 2100.00           |

these have the same information as the original table but are just more optimized.